IRONHACK • DATA SCIENCE & MACHINE LEARNING

Can Machine Learning Predict Fuel-Cell Voltage?

A reproducible workflow using PEM fuel-cell polarization data

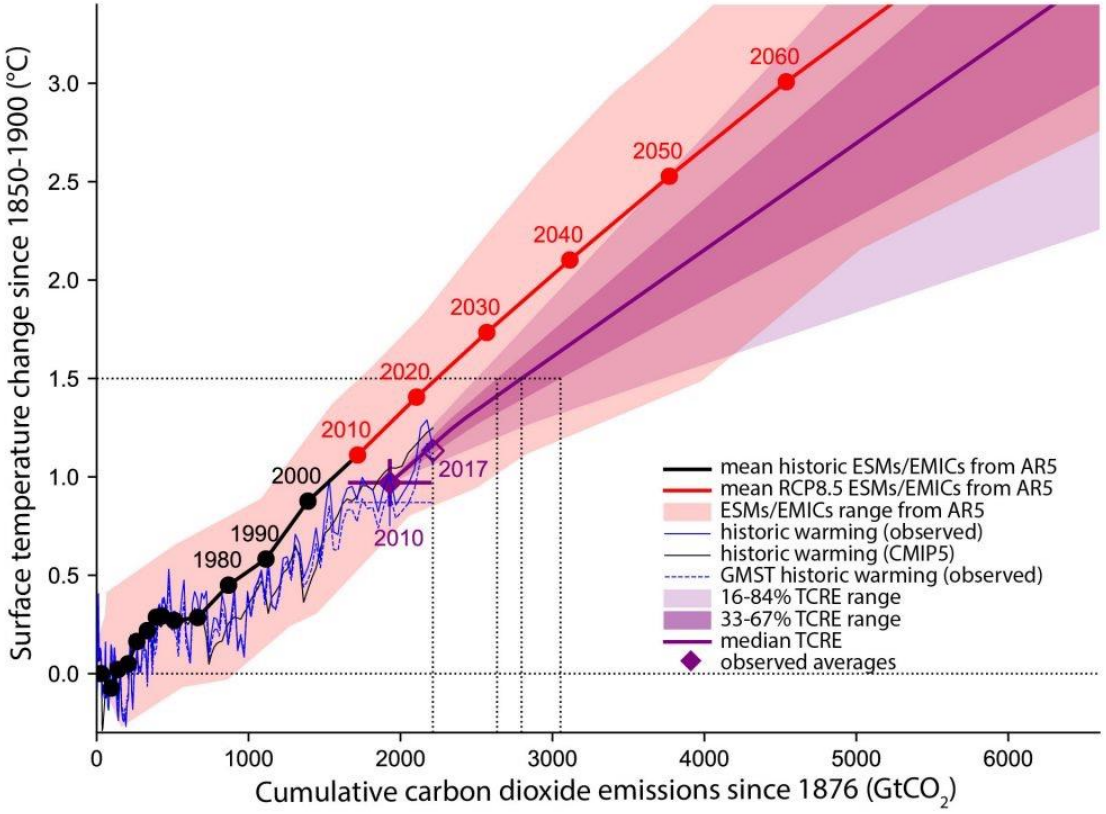
Project 2 | Machine Learning: From Data to Predictions

Theophilus Kobina Sarpey

Supervised learning • Grouped validation • Error analysis

Why Renewables Research? Global Warming – An Existential Threat

[I]



[II]



[V]



[III]



[VI]



→ **NOT** Limiting Global Warming = **Bad Time**

[IV]



[I] Rogelj, J., D. et al., 2018

[II] https://therevelator.org/climate-change-extinction/

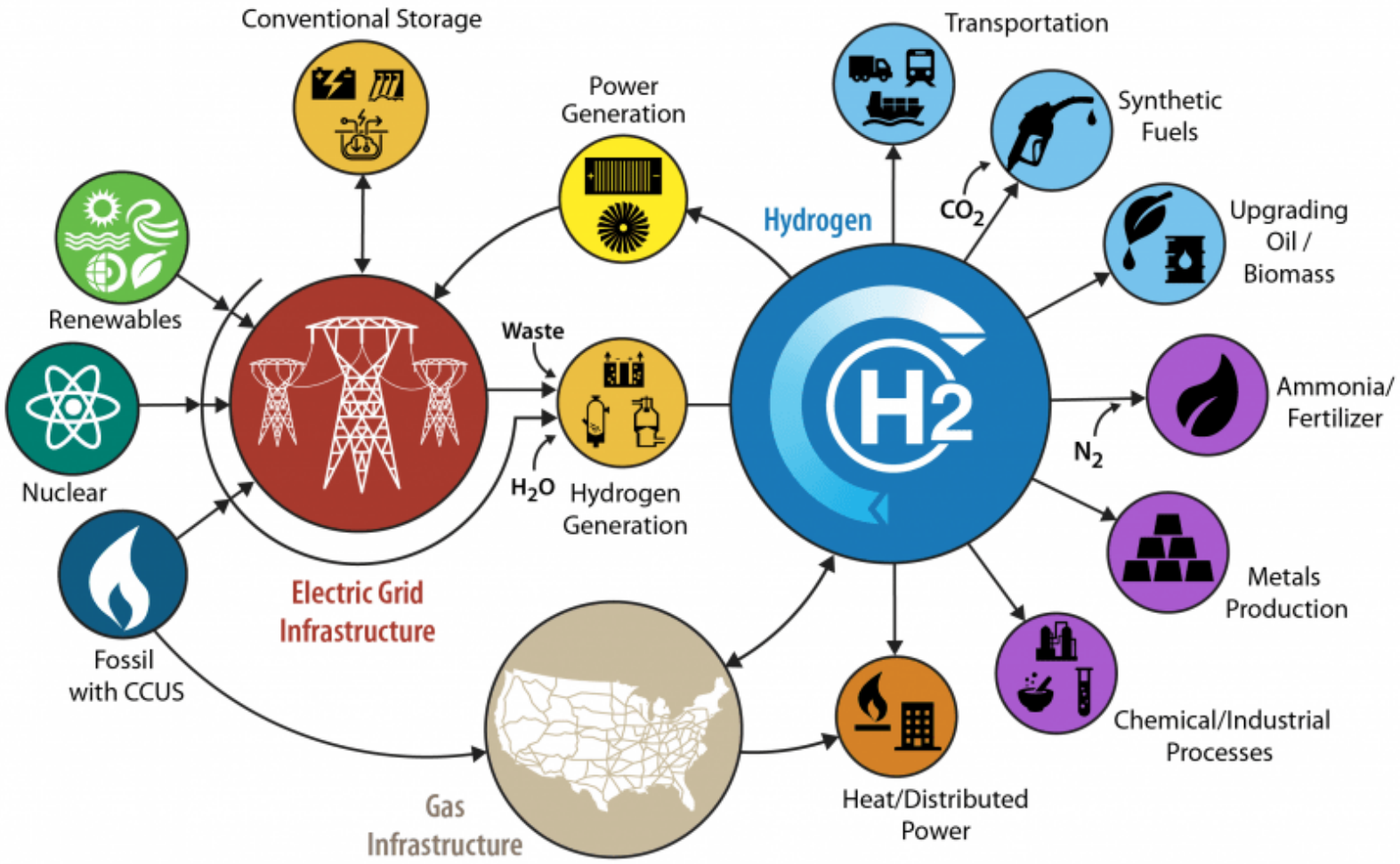
[II] https://www.thedailystar.net/country/impact-of-climate-change-across-coastal-bangladesh-induced-salinity-affecting-soil-1688872

[IV] https://edition.cnn.com/europe/live-news/deadly-flooding-in-europe-07-15-21/h_265055052eac25b457d0aaeead38b4a3

[V] https://phys.org/news/2018-10-woes-drought-germany-rhine-shallow.html

[VI] D. Moore,Saving the planet, Mexican Journal of Biotechnology, 2021

The central and significant role of Hydrogen



https://www.energy.gov/eere/fuelcells/h2scale

Fuel Cells (FCEVs): Applications in the Transport Sector

Cars










Buses





Trucks




Logistics



Ships




Trains



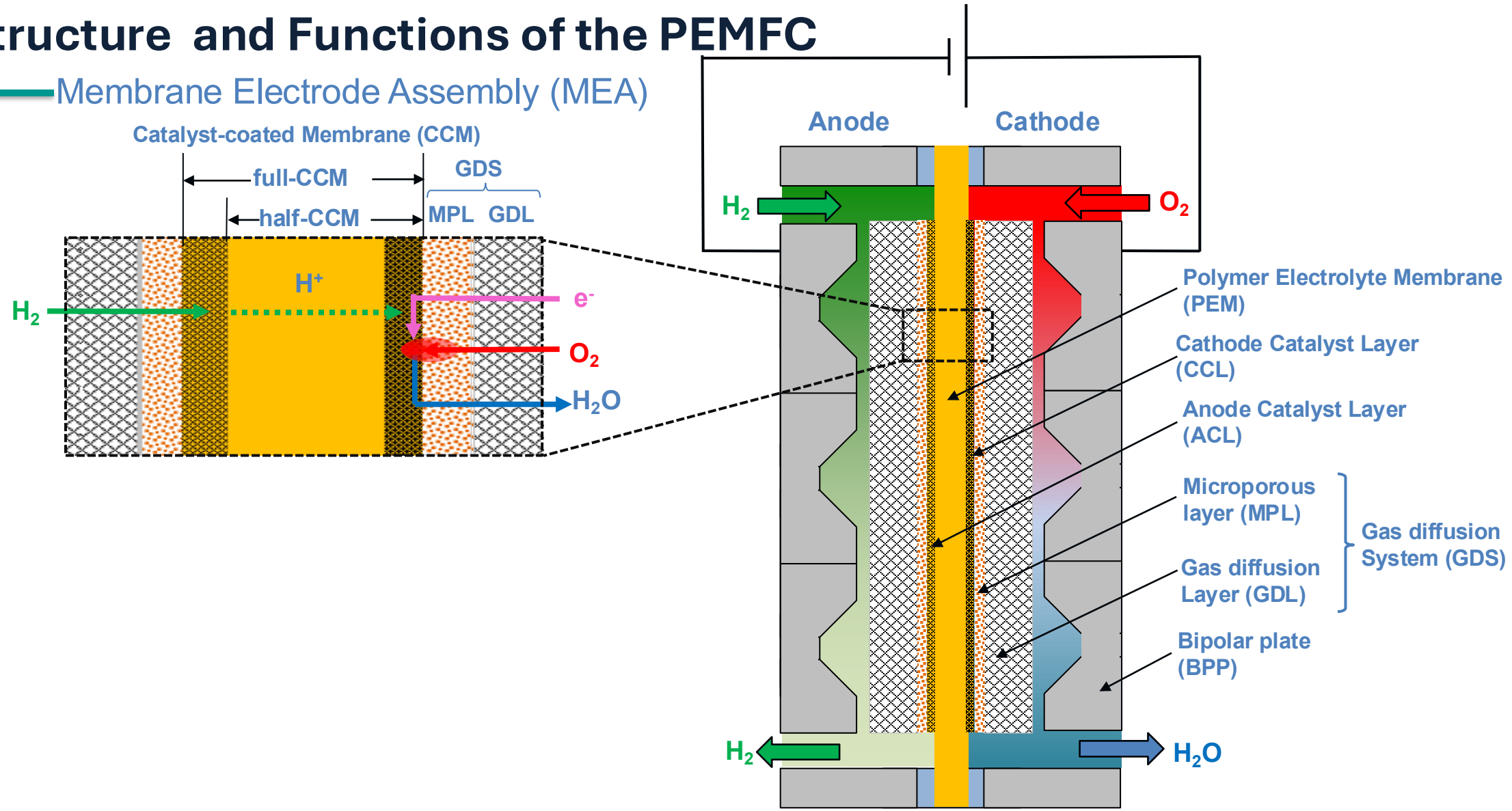
Airplanes



Source: Daimler, Honda, VW, Hyundai, GM/Opel, FEV, APTS, Proton Motor, FAUN, Airbus, DLR, HDW, Still, Linde, Alstom

Structure and Functions of the PEMFC

Membrane Electrode Assembly (MEA)



Why predict fuel-cell voltage?

Voltage

A key signal of how a PEM fuel cell responds to operating conditions.

Experiments

Measurements take time and vary across test protocols and source files.

ML screening

A model may help estimate expected voltage and prioritize follow-up measurements.

Research question

Can a supervised model predict cell voltage from operating conditions and experimental context when entire source files are held out for testing?

Scope: predictive screening on a specific dataset; not a substitute for electrochemical testing or a mechanistic model.

Dataset: many measurements, multiple protocols

144 files
Files in the source archive

26 files
Polarization-curve files selected

9 families
Distinct test-family labels

5,005 rows
Usable observations after processing

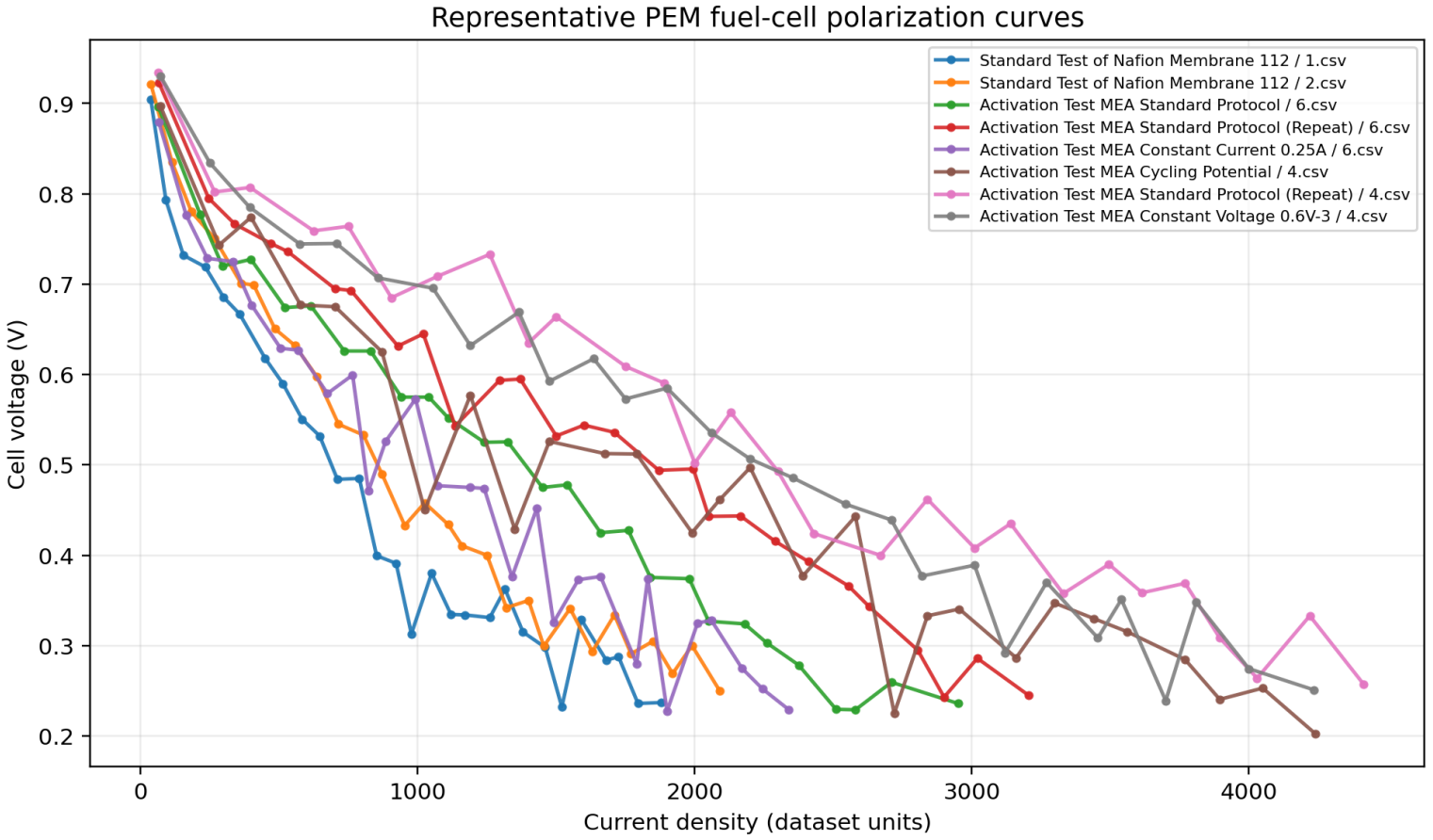
- Source: Hamidi, Haghighi & Askari, ChemRxiv (2020), DOI: 10.26434/chemrxiv.11902023.
- Project materials identify the dataset as CC BY 4.0 License.
- Selected task: predict cell voltage from polarization-curve records.
- Derived power density was excluded as a predictor to avoid target leakage.

Why it matters - Splitting decision!

- Measurements (rows) from the same source file are related.

→ A random row split could make testing look easier than it really is.

What does a polarization experiment look like?

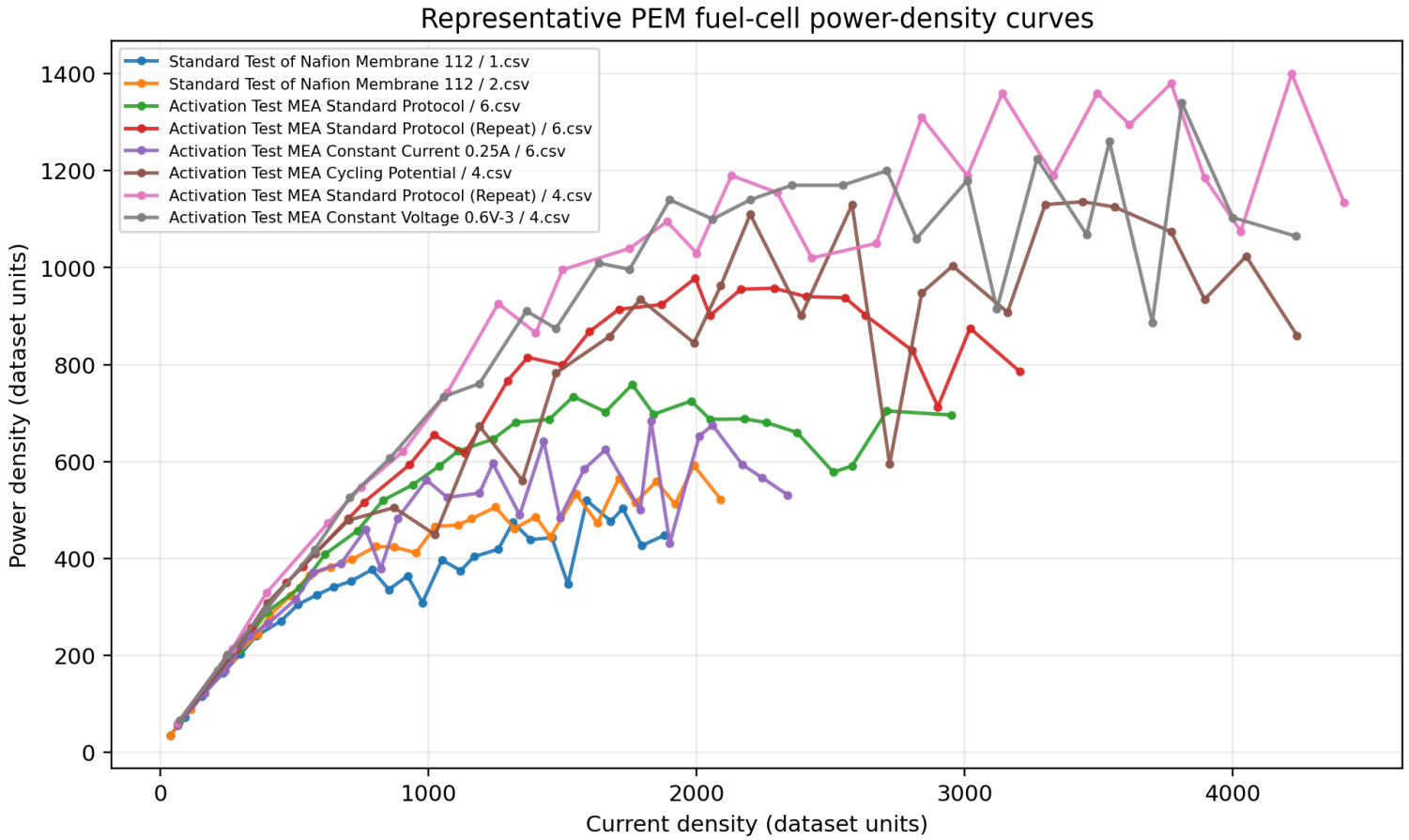


Rrepresentative polarization curves from the processed project dataset.

What stands out in the curve features

- Voltage typically changes as current density increases.
- Voltage declines as current density increases.
- Curves differ across experiments and protocols.
- The model learns patterns in measured data; it does not prove a physical mechanism.

From Polarization to Power Density

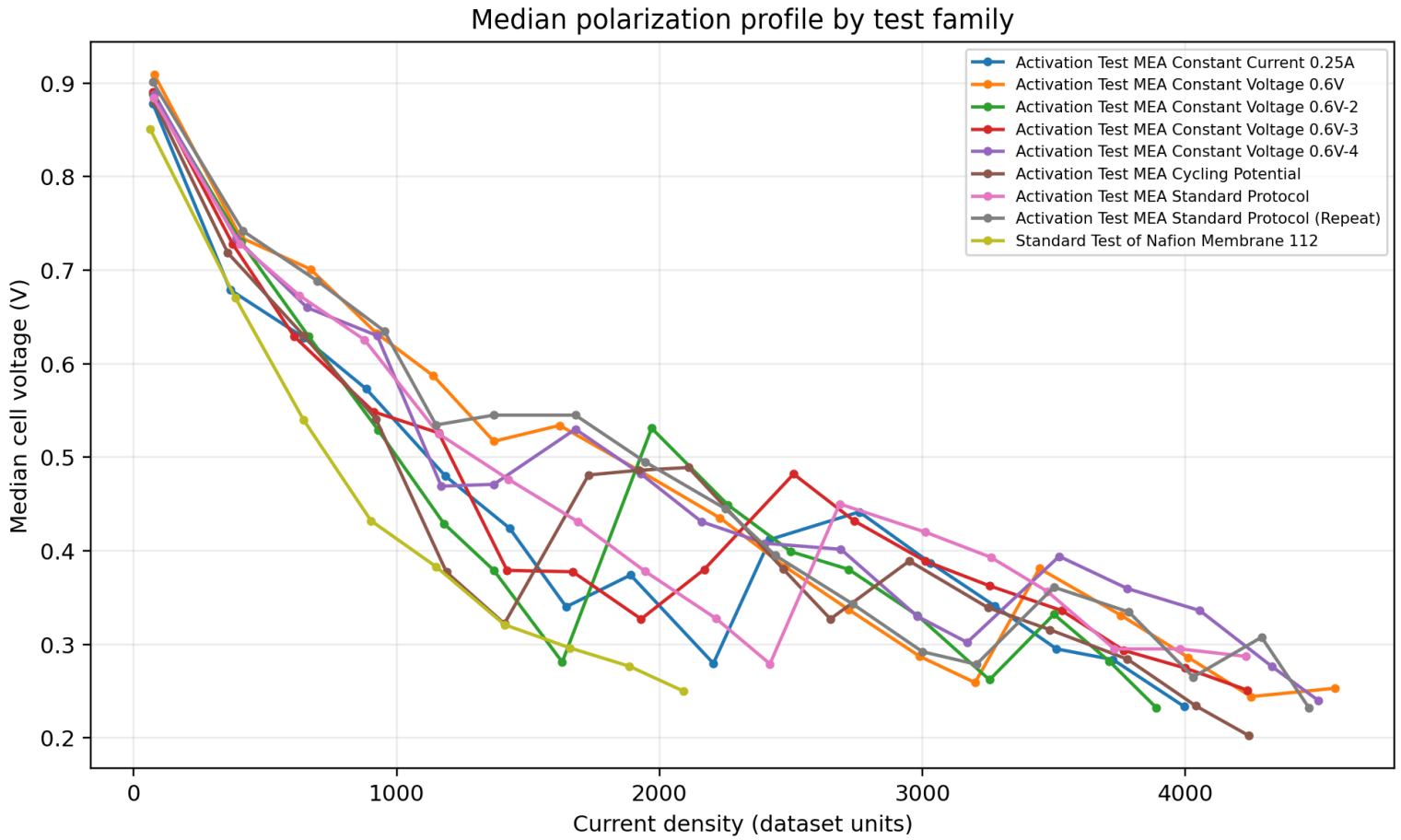


Rrepresentative polarization curves from the processed project dataset.

The curve features

- Power density depends on both current density and cell voltage.
- It typically reaches a maximum before declining.
- Curves differ across experiments and protocols.
- The model learns patterns in measured data; it does not prove a physical mechanism.

Protocol differences are part of the problem



Interpretation: protocol variation is visible in the dataset; it is not necessarily measurement error.

Key implication

A model may learn both operating-condition relationships and differences associated with test families.

Generalization to a new laboratory or MEA must be tested separately.

Validation designed to reduce leakage

20 files

Training partition • 3,692 rows
73.8%

6 files

Held-out test • 1,313 rows
26.2%

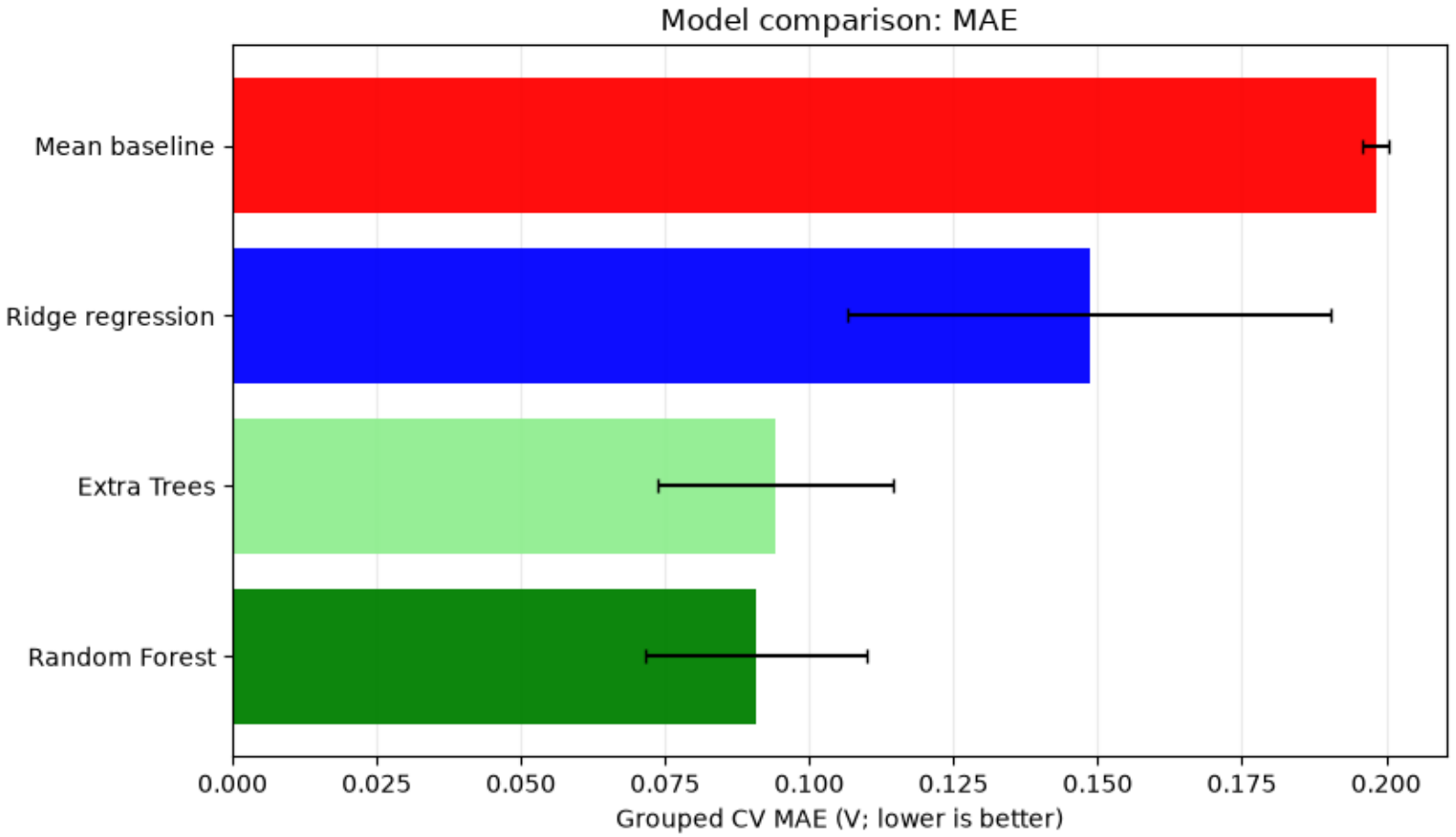
0 overlap

No source-file overlap between train and test

- Instead of a simple random split of individual rows, grouped validation was preferred.
- Model selection used 3-fold GroupKFold with source-file groups.
- Hyperparameter search used training data only.
- The held-out test partition was reserved for final evaluation.
- Related protocols may still appear in both partitions, so this is not external-laboratory validation.

Plain-language version: train on some experimental files; test on entirely different files.

Model comparison: tree models performed best



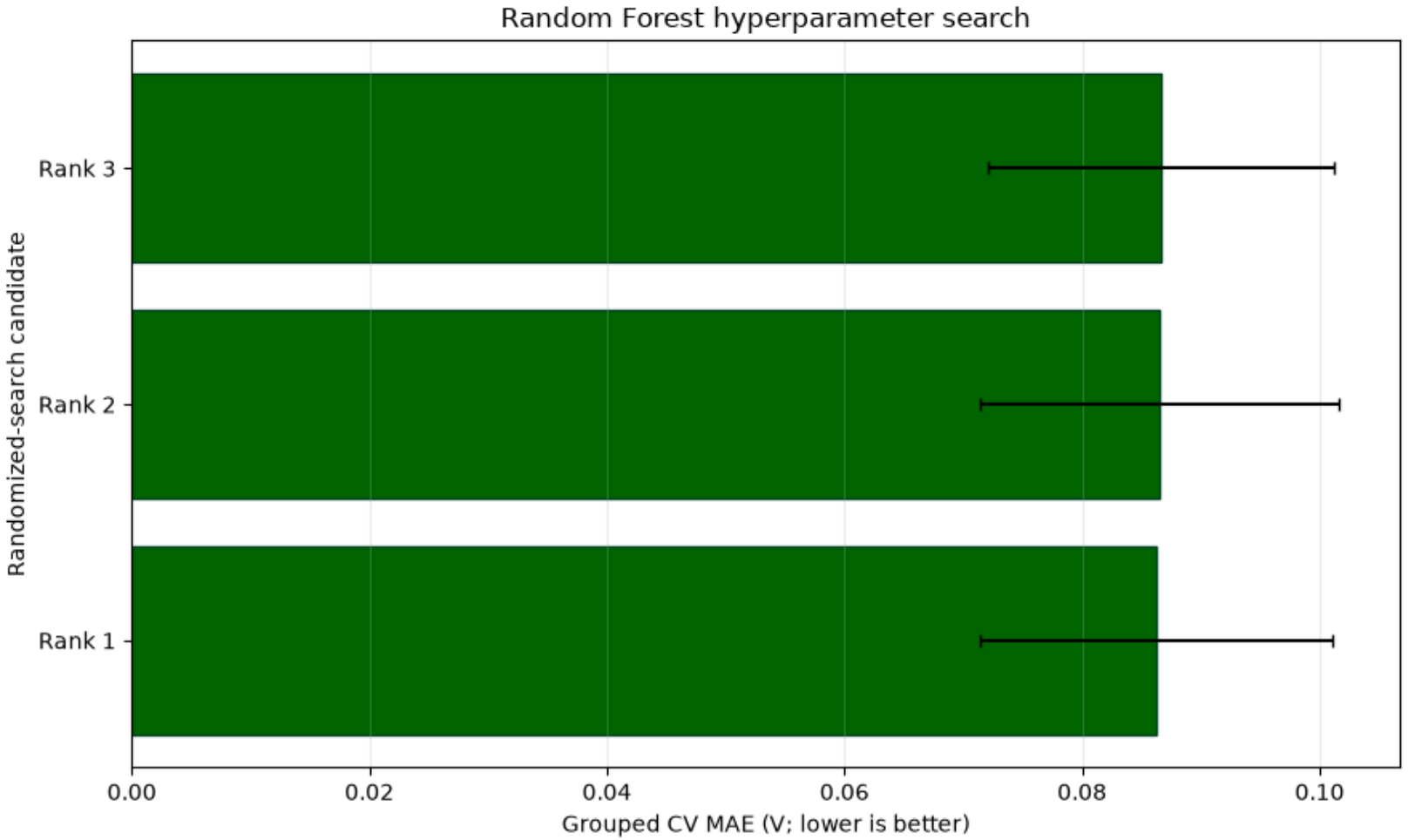
All models compared using the same grouped cross-validation strategy.

Grouped-CV MAE

Random Forest	0.0851 ± 0.0099 V
Extra Trees	0.0889 ± 0.0122 V
Ridge	0.1437 ± 0.0045 V
Mean baseline	0.1982 ± 0.0012 V

Lower MAE is better. Values are mean ± fold standard deviation.

Tuning selected a Random Forest pipeline



Tuning scores are selection estimates; the final held-out test is reported separately.

Selected settings

- 60 trees
- Maximum depth: 14
- Minimum samples per leaf: 4
- Max features: 0.8

Best grouped-CV MAE

0.08117 V

Final test results: tuned model beat the baseline

0.19650 V

Mean baseline • test MAE

0.06589 V

Tuned Random Forest • test MAE

Baseline test metrics

RMSE 0.22708 V
R^2 −0.00087

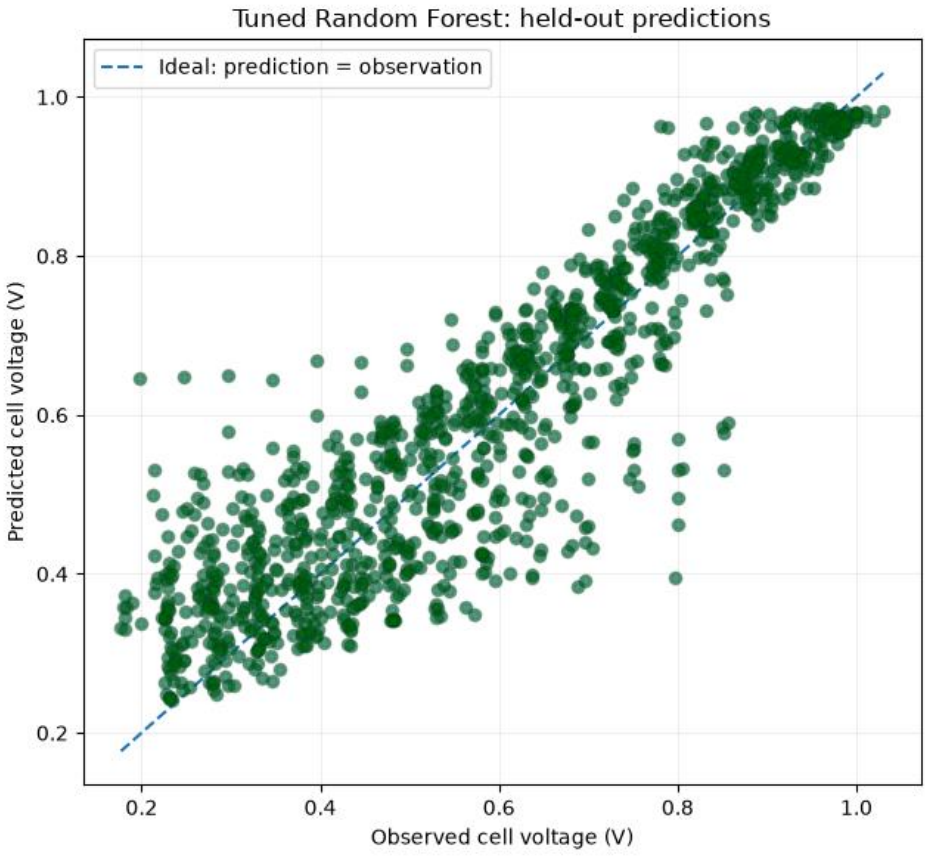
Tuned Random Forest test metrics

RMSE 0.09005 V
R^2 0.84259

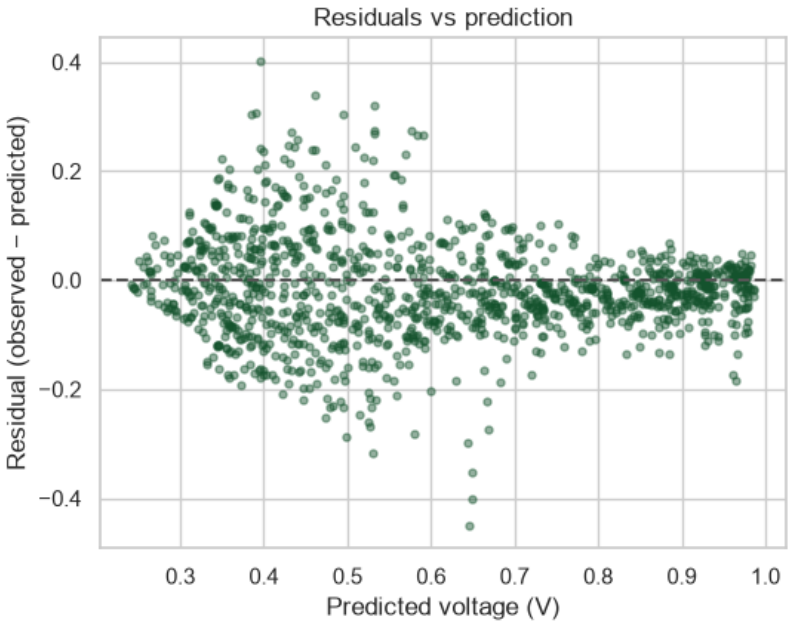
≈ 66.5% lower MAE than the mean baseline on the held-out test set

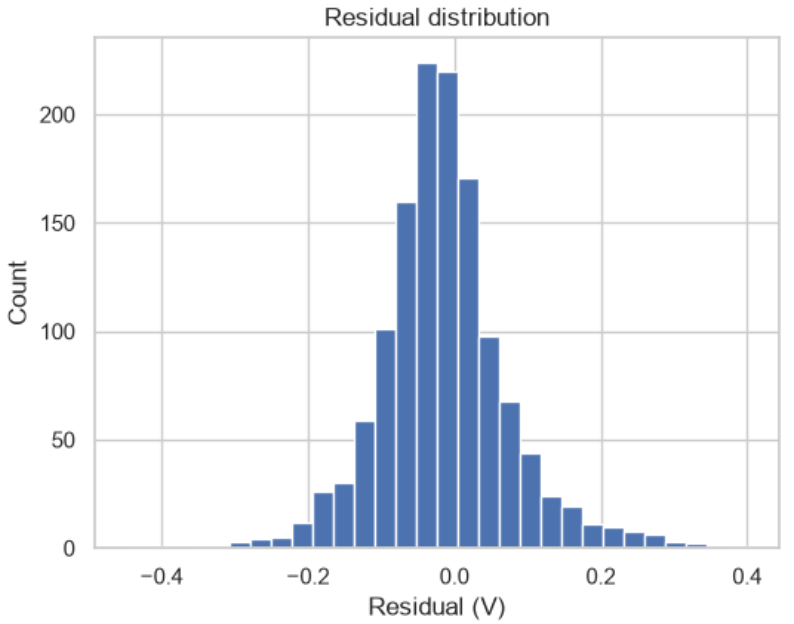
Metrics are from the saved final_test_metrics.csv; do not confuse (mix) them with cross-validation scores.

Prediction quality and residual diagnostics



Predicted vs observed voltage

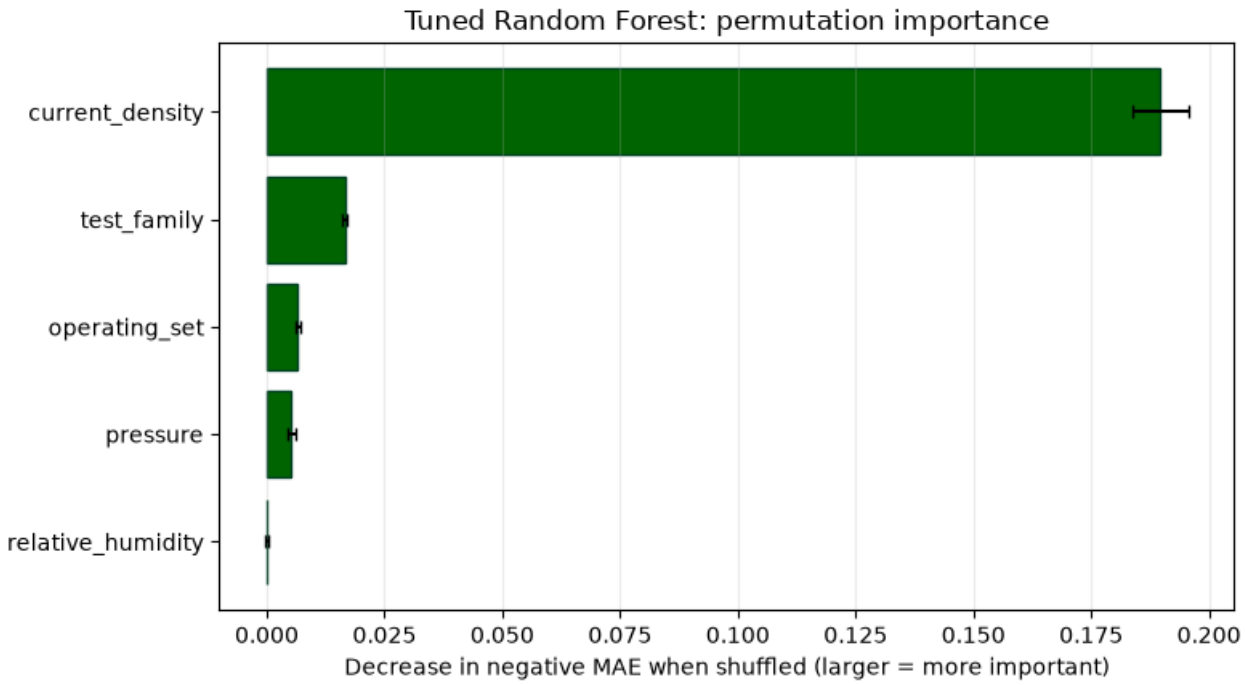


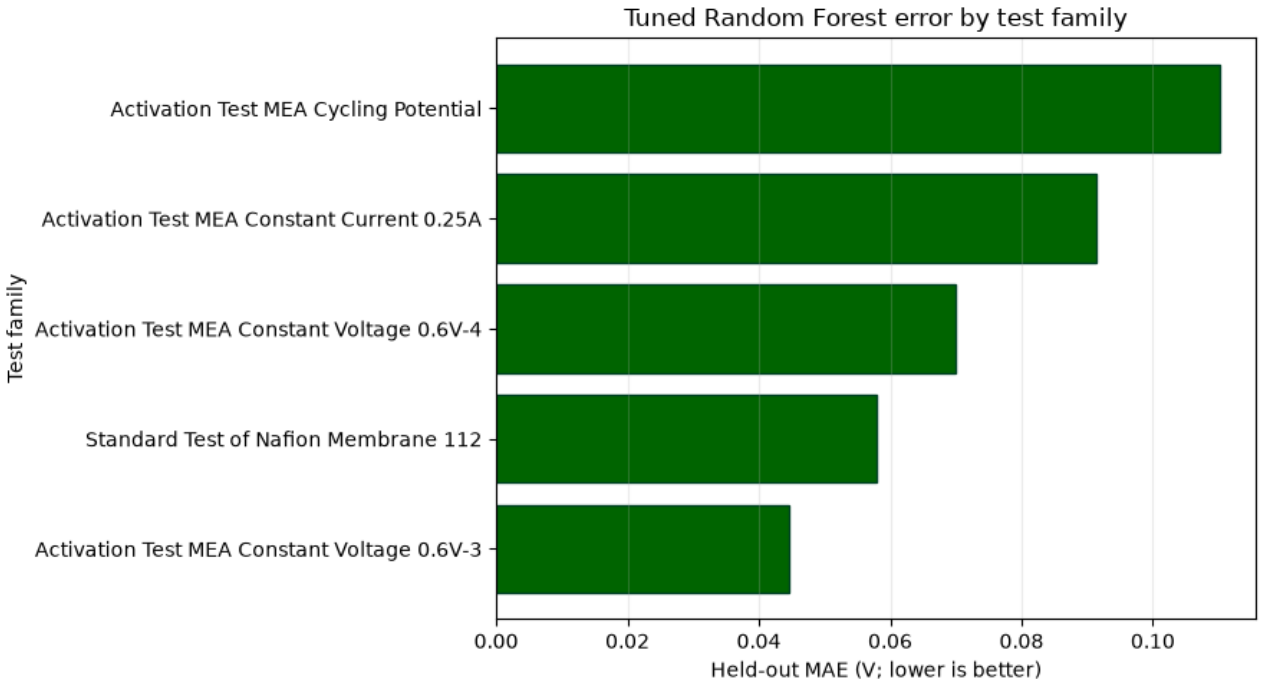


Residual patterns and spread

Inspect systematic deviations, not only a single summary score.

What matters—and where errors increase





Permutation importance: predictive utility, not causality.

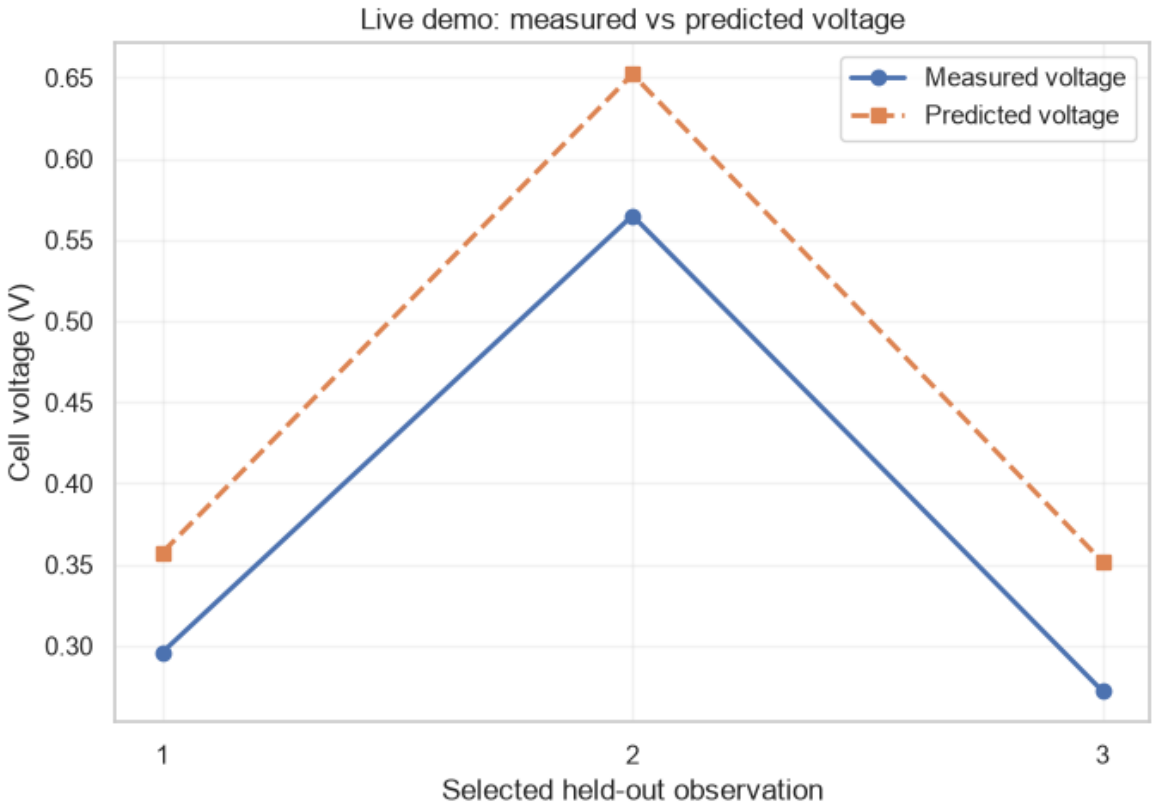
Error varies by test family; inspect segments before deployment.

Live demo: from inputs to predicted voltage

- Load the saved pipeline (`best_model.joblib`).
- Provide the required input features for a few rows.
- Generate predictions and compare with recorded voltage where available.
- A prediction is conditional on the training data and feature definitions.

Demo checklist

- Fresh kernel / environment ready
- Model file loads successfully
- Feature names and units checked
- Saved prediction output available



Conclusion: promising screening, not universal validation

- A reproducible pipeline processed 5,005 observations from 26 polarization-curve files.
- Grouped validation made the test more realistic than a random row split.
- The tuned Random Forest achieved 0.06589 V MAE on held-out source files.
- Performance may change for new labs, MEAs, or operating regimes.

Next steps

- External validation on independent experiments.
- Define an acceptable voltage-error tolerance.
- Document feature availability before real-world use.

Main lesson: reliable machine learning depends on how data are split, compared, and interpreted—not just which model wins.

Thank you • Questions?

Model selection – Grouped cross-validation results

